# Factorization and Topological States in \$c=1\$ Matter Coupled to 2-D Gravity

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**Abstract**

Factorization of the \$N\$-point amplitudes in two-dimensional \$c=1\$ quantum gravity is understood in terms of short-distance singularities arising from the operator product expansion of vertex operators after the Liouville zero mode integration. Apart from the tachyon states, there are infinitely many topological states contributing to the intermediate states.


## 1 Source Metadata

This sample is based on arXiv record `hep-th/9108027`. Authors: Norisuke Sakai, Yoshiaki Tanii.

## 2 Extracted Prose 1

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## 3 Extracted Prose 2

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